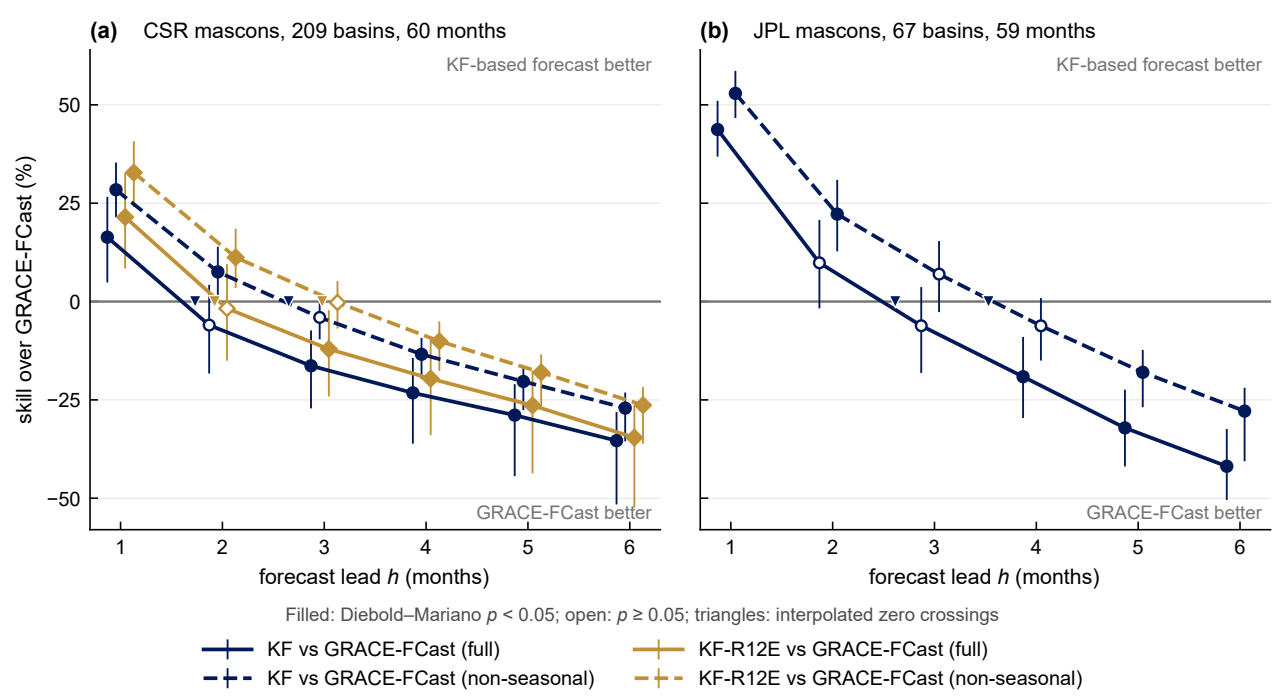

(a) CSR mascons, 209 basins, 60 months
KF-based forecast better
GRACE-FCast better
skill over GRACE-FCast (%)
50
25
0
−25
−50
1
2
3
4
5
6
forecast lead h (months)
(b) JPL mascons, 67 basins, 59 months
KF-based forecast better
GRACE-FCast better
forecast lead h (months)
Filled: Diebold–Mariano p < 0.05; open: p ≥ 0.05; triangles: interpolated zero crossings
KF vs GRACE-FCast (full)
KF vs GRACE-FCast (non-seasonal)
KF-R12E vs GRACE-FCast (full)
KF-R12E vs GRACE-FCast (non-seasonal)